

SOFTWARE ENGINEER (BACKEND / SYSTEMS)

Backend-focused Software Engineer with experience in **distributed systems, REST APIs, and cloud infrastructure**. Built scalable applications and **AI-powered systems (RAG, LLMs)** using modern backend frameworks.

TECHNICAL SKILLS

Languages	: C, C++, Java, Python, Go, JavaScript
Backend Systems	: REST APIs, Distributed Systems, Microservices, Concurrency, Multithreading, Memory Management
Frameworks	: Spring Boot, FastAPI, Node.js, Express, React.js, Next.js
Databases	: PostgreSQL, MySQL, MongoDB, Redis (Caching, Indexing)
Cloud & DevOps	: AWS (EC2, S3, Lambda), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), ArgoCD
Tools	: Git, Linux, Prometheus, Grafana, QEMU, Wireshark
AI/ML	: LLM, RAG, Embeddings, Vector Databases (ChromaDB), Claude Code, Codex

EXPERIENCE

Teaching Assistant - System Foundations & Computer Architecture <i>University of San Francisco</i>	August 2025 – Present San Francisco, CA
- Guided 100+ students in understanding core concepts of Operating Systems and Computer Architecture, including processes, memory management, and concurrency - Led weekly lab sessions and debugging support, helping students resolve low-level issues in C/Assembly such as segmentation faults and memory leaks	
Software Engineer Intern <i>American Bureau of Shipping</i>	January 2023 – June 2023 Maharashtra, India
- Delivered a React + Spring Boot performance management platform serving 400+ employees, reducing manual HR processing time by 35% through automated workflows - Accelerated report generation by 40% by redesigning Oracle DB – IBM Maximo synchronization via optimized REST APIs, eliminating a recurring bottleneck	

EDUCATION

University of San Francisco <i>Master of Science in Computer Science</i>	San Francisco, CA, USA August 2024 – May 2026
Shivaji University <i>Bachelor's Of Technology in Computer Engineering</i>	Sangli, MH, India July 2019 – July 2023

PROJECTS

Distributed MapReduce Engine - Built a distributed MapReduce engine in Go from scratch on top of a custom DFS, orchestrating map and reduce tasks across a 12-node HPC cluster with plugin-based job binaries and configurable reducer counts - Implemented full shuffle pipeline using FNV-32a hash partitioning and k-way min-heap merge across sorted mapper streams, following the Google MapReduce design	<i>Go, Protocol Buffers, Distributed Systems</i>	Source Code
Magnet Arena (Cloud-Native Game) - Provisioned production AWS infrastructure (EKS, RDS, VPC) exclusively via Terraform IaC and engineered a 4-stage Git-driven CI/CD pipeline (Dev → QA → UAT → Prod) with zero-downtime Blue/Green deployments on Kubernetes. - Deployed self-hosted Prometheus + Grafana observability stack inside EKS with OAuth2-secured access, cluster-wide resource alerting, and centralized log aggregation via Loki across all microservices.	<i>AWS EKS, Terraform, Kubernetes</i>	Source Code
ClipStudy (AI-Powered Video Learning Platform) - Architected serverless FastAPI backend on AWS Lambda with SAM/IaC template, integrated VPC, RDS MySQL, and Cognito authentication to handle distributed video processing without managing servers or worrying about capacity scaling. - Designed asynchronous job queue using SQS + EC2 worker polling, enabling GPU-accelerated transcription and LLM-generated flashcards while maintaining sub-second API response times for 10+ concurrent user submissions.	<i>React, FastAPI, AWS</i>	Source Code
EasyShare (AI-Powered Bill Splitting & File Sharing) - Engineered AI receipt analysis pipeline using OpenAI vision API with Zod schema validation, post-processing, and price normalization; automatically extracts restaurant details and itemizes bills for instant splitting. - Developed real-time collaboration features with Socket.IO, Redis-backed rooms, and Clerk auth enabling concurrent file sharing, messaging, and bill management across participants with fine-grained access control.	<i>Next.js, Web Sockets, Redis, LLM API</i>	Source Code
Vama Collections - Architected a production retail POS system for a real-world shop, exposing 9+ REST endpoints managing inventory, purchase orders, billing, and tax reporting; implemented a tax calculation engine with multi-rate support, automatic rounding, and discount logic. - Designed domain-driven architecture with 8 aggregate roots across JPA entities, Specifications-based filtering, transaction management, and Flyway migrations; added unit tests for core billing logic ensuring accurate financial calculations.	<i>Java, Spring Boot, PostgreSQL</i>	Source Code